Contribution statement

The work was divided equally: each author led one third of the project, and all three contributed equally overall. All authors shared the research question, the reading of the literature, the interpretation of results, and the writing and review of the extended abstract.

Emmanuel Azubuike (2698752): framing analysis (H2)

- Built the embedding pipeline (fastText) and ran the hyper-parameter search for seed stability on held-out years (Table S8), including the comparison with word2vec.
- Implemented the profile-shift and neighbour-overlap measures, the permutation null and the frequency-matched word sets for framing.
- Designed the directional anchor test and its robustness checks: windows 2 and 5, joined word forms, and the tests without *land* and without the anchor *endlu*.
- Derived the detection floor and carried out the word-sense analysis of *umhlaba*.

George Ayensu (2722188): data and salience analysis (H1)

- Loaded and cleaned both newspapers' transcriptions and built the date- and size-matched samples.
- Ran the exploratory classifier and yearly series that led to the religion concept; built and checked the concept word stems, including correcting the *law* stem.
- Computed salience rate ratios with bootstrap confidence intervals, the matched-set null and within-paper era drift.
- Built and hand-checked the readers' letter detector, and ran the letters-versus-rest and same-editor contrasts that triangulate control against editor and time.

Banele Mkhize (2663814): tokenizer fairness (H3), responsible NLP and reproducibility

- Designed and ran the tokenizer cost experiments for XLM-R, mBERT and Serengeti on modern, spelling-altered and historical text, and the tokenizer vocabulary analysis.
- Led the responsible-NLP analysis (speaker review, extractive readings, spelling normalisation) and completed the NeurIPS checklist.
- Produced the figures and supplementary material, organised the code, `README.txt` and `requirements.txt`, and checked that every reported number is regenerated by the submitted code.

All authors reviewed and approved the final report and code, and take joint responsibility for them.